

Original Article

Similarity-based profiling of pharmacovigilance data for drug safety pattern discovery

Seongjae Park¹, SeongJin Wi², Heeseon Jo¹, Yuseong Kwon¹, Nakjung Jeon^{1,2,3,*}, and Haeseung Lee^{1,*}

¹College of Pharmacy and Research Institute for Drug Development, Pusan National University, Busan 46241, Korea, ²Graduate School of Data Science, Pusan National University, Busan 46241, Korea, ³Biomedical Research Institute, Pusan National University Hospital, Busan 49241, Korea

ARTICLE INFO

Received March 20, 2026
Revised April 3, 2026
Accepted April 6, 2026
Published online May 7, 2026

*Correspondence

Haeseung Lee
E-mail: haeseung@pusan.ac.kr
<https://orcid.org/0000-0002-9947-4032>
Nakjung Jeon
E-mail: nakjung.jeon@pusan.ac.kr
<https://orcid.org/0000-0001-5500-8841>

Key Words

Drug-related side effects and adverse reactions
Korea Adverse Event Reporting System
Pharmacovigilance

ABSTRACT Spontaneous adverse event (AE) reporting systems enable large-scale pharmacovigilance but are typically analyzed as isolated drug–event pairs. Here, we constructed a drug–drug similarity network at an ingredient-level, using 968,966 reports from the Korea Adverse Event Reporting System (KAERS; 2020–2024). After preprocessing, 382,530 reports involving 1,058 ingredients and 3,749 AE events in MedDRA preferred terms (PTs) were retained. Significant ingredient–event signals were identified using proportional reporting ratio, reporting odds ratio, χ^2 , and information component. Pairwise similarity between ingredients was quantified using a hypergeometric test based on shared significant PTs with a false discovery rate ingredients and 1,111 PTs, resulting in a network of 150 ingredients and 1,267 edges. Community detection revealed modules that recapitulated known pharmacological classes, including antineoplastic agents and contrast media, and exhibited clinically coherent safety profiles. Notably, cross-class clustering, including statins with anti-infective and anti-inflammatory agents, suggested shared downstream biological effects beyond primary indications. These findings demonstrate that a signal-based drug similarity network derived from spontaneously reported data can capture clinically meaningful safety patterns and reveal latent relationships across therapeutic classes, thereby providing a scalable approach to pharmacovigilance and hypothesis generation.

INTRODUCTION

Post-marketing pharmacovigilance often utilizes spontaneous adverse event (AE) reporting systems, which are critical for capturing rare and unexpected events that may not emerge during pre-approval clinical trials [1]. However, the AE reports accumulated in such systems are inherently limited by under-reporting, heterogeneous data quality, and the absence of explicit exposure denominators, which preclude the calculation of true incidence rates or the establishment of definitive causal inferences [2]. Disproportionality analysis partially addresses these challenges by identifying specific drug–event pairs that are reported more

frequently than expected relative to the background reporting distribution of the entire database [3]. Within this framework, a safety signal is typically defined as the disproportionate reporting of an AE for a specific drug compared to its reporting frequency across all other agents.

Traditional pharmacovigilance tends to evaluate these signals in isolation or within rigid therapeutic classes by using disproportionality measures [4–14]. However, this approach may overlook the fact that AEs frequently cluster across disparate agents that share pharmacological mechanisms, therapeutic indications, or off-target liabilities. Network representations provide a robust formalism to integrate these patterns by characterizing each drug



This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License, which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited.
Copyright © Korean J Physiol Pharmacol, pISSN 1226-4512, eISSN 2093-3827

Author contributions: S.P. conducted the overall analyses and drafted the manuscript. S.W. mapped Anatomical Therapeutic Chemical (ATC) codes for each ingredient. H.J. and Y.K. standardized ingredient names and mapped drug records to unique ingredients. N.J. and H.L. conceived and led the overall study and contributed to manuscript writing and revision. All authors reviewed the manuscript and approved the final version.

as a vector of statistically supported signals [15]. By leveraging the shared signal topography between ingredients, this network-based approach transcends the limitations of univariate drug-event analysis, allowing the identification of safety clusters, where co-occurring signals provide stronger evidence of an underlying pharmacological mechanism than a single isolated association.

In this study, we utilized the Korea Adverse Event Reporting System (KAERS) to construct a comprehensive, ingredient-level drug–drug network. The KAERS provides a national-scale corpus of spontaneous reports, which, when coupled with structured ingredient identifiers and MedDRA-coded AEs, enables systematic data mining of post-marketing safety profiles [16]. Through the application of community detection algorithms, we identified distinct modules of ingredients with shared AE signatures. These module-level phenotypes establish a data-driven framework for signal prioritization and facilitate the systematic generation of drug-repurposing hypotheses by elucidating shared toxicological pathways across disparate therapeutic classes.

METHODS

Data source

This study utilized spontaneous AE reports from the KAERS curated by the Korea Institute of Drug Safety and Risk Management (KIDS) for the period 2020–2024 (Application No: 2603A0001). For this analysis, we integrated five core tables from the KAERS database: demographic (DEMO), drug information (DRUG, DRUG1, and DRUG2), and AE, which were linked using a unique report identifier. The DEMO table provides case-level metadata, including patient (*e.g.*, age and sex) and report characteristics. Within the DRUG tables, drug records were systematically encoded using the Ministry of Food and Drug Safety (MFDS) ingredient code system (M-code) and included within-report sequence identifiers to enable accurate linkage to specific drug exposures. Lastly, within the AE table, specific AEs were coded utilizing the Medical Dictionary for Regulatory Activities (MedDRA) version 28.0 at the lowest level term (LLT) hierarchy.

Data preprocessing

A KAERS report may include multiple drug records, each identified by a within-report sequence number and mapped to one or more ingredient identifiers. To focus on drug exposures most relevant to the reported events, we retained only drug records coded as “suspected” or “interacting” with respect to the AE; drug records coded as “concomitant” or “unadministered” were excluded from the primary analyses. To focus on systemic use, topical and external drug records were excluded based on the formulation and route of administration. Vaccines were also excluded because of their unique AE profiles and surveillance practices, which are

primarily monitored by the Korea Disease Control and Prevention Agency (KDCA), whereas other drugs are regulated by the MFDS. In addition, we excluded multi-ingredient drug records in which a single drug record mapped to two or more ingredient identifiers ($\text{INGR_SEQ} \geq 2$) to mitigate ambiguity in ingredient attribution. Reports containing multiple single-ingredient drug records were eligible.

AEs in the KAERS database are originally recorded at the LLT level. LLTs were mapped to their corresponding preferred terms (PTs), and all analyses were conducted at the PT level to reduce semantic redundancy across synonymous LLT expressions. Throughout this study, the term “AE” refers to PT-level events unless otherwise specified. However, to provide a more descriptive and clinically intuitive representation of the identified patterns, results are presented using the original LLTs associated with each signal.

Reports with implausible age values (> 200 years) were excluded. Reports with missing drug or AE information as well as those containing masked ingredient codes (MSK) were excluded to ensure that ingredient–AE pairing was possible. Data were deduplicated by selecting only the primary report per case to avoid redundant signal detection. To address nomenclature heterogeneity, ingredient names were standardized to a single generic identifier, regardless of salt form, strength/concentration, and dosage form descriptors.

Disproportionality analysis and safety signal definition

For each ingredient–AE pair, we constructed a 2×2 contingency table to calculate the proportional reporting ratio (PRR), reporting odds ratio (ROR), χ^2 statistics, and the information component (IC) using a Bayesian Confidence Propagation Neural Network [3]. At the ingredient level, an AE (*i.e.*, PT) was formally identified as a safety signal only if it concurrently satisfied all of the following criteria: number of reports ≥ 3 , $\text{PRR} \geq 2$, $\text{ROR} \geq 2$, $\chi^2 \geq 4$, and $\text{IC}_{025} \geq 0$. While safety signals were identified at the PT level to ensure statistical robustness, findings are presented at the LLT level to preserve the clinical verbatim and reflect specific manifestations of the reported AE. The seriousness of each signal was determined based on the event-level result codes inherent in the KAERS. Serious AEs (SAEs) resulting in death, disability, or hospitalization, were analyzed separately as high-risk signals.

Network construction

To account for random co-occurrence, a hypergeometric test was applied to all ingredient pairs that exhibited at least one safety signal. The test evaluates whether the probability that the number of shared safety signals between ingredients is higher than that expected by chance, based on the total number of safety signals and the frequency of shared safety signals for each ingredient.

This procedure was intended to downweight shared events that are broadly common across many ingredients and are therefore less informative for defining specific ingredient similarities.

p-values derived from the hypergeometric tests were adjusted using the Benjamini–Hochberg procedure to control the false discovery rate (FDR). The FDR threshold was set at 1×10^{-4} as a conservative measure to minimize Type I errors in the high-dimensional screening of ingredient pairs. In this network, each node represents an ingredient, and an edge between two nodes indicates a significant pair of ingredients sharing safety signals exceeding the specified thresholds. To determine the optimal frequency threshold for shared safety signals, a data-driven optimization process was conducted to identify the inflection point (elbow) in network density [17]. This metric, calculated as the ratio of actual edges to all possible edges within the network, was monitored across varying thresholds to maximize safety signal specificity while preserving sufficient connectivity for robust and stable network analysis. Based on these criteria, the resulting network consisted only of ingredient pairs that satisfied the predefined statistical and frequency thresholds.

Community detection and annotation

Each ingredient was represented as a binary vector over the complete AE sets (i.e., PT) identified in the analytic dataset, where

a value of 1 indicated that the AE satisfied the predefined criteria for a safety signal and 0 otherwise. This representation resulted in an ingredient-by-PT matrix that captured the unique AE signature of each ingredient in a standardized feature space. Based on this representation, the community structure was identified using the Louvain algorithm [18], which iteratively optimizes modularity to detect densely connected groups of nodes and is well-suited for the analysis of large, sparse graphs.

Each module was annotated by summarizing two complementary features: (i) the distribution of anatomical therapeutic chemical (ATC) second-level categories among the constituent ingredients and (ii) representative PTs, defined according to the number of ingredients within the module exhibiting the corresponding significant signal. For ingredients assigned to multiple ATC codes, a single representative ATC second-level category was selected for visualization based on the most frequently observed second-level category for that ingredient in the dataset.

Software

Data preprocessing and statistical analyses were performed using Python (version 3.10.12). Network construction and community detection were implemented using standard graph-analysis libraries in Python, and network visualization was performed using Cytoscape (version 3.10.4). All non-network figures were

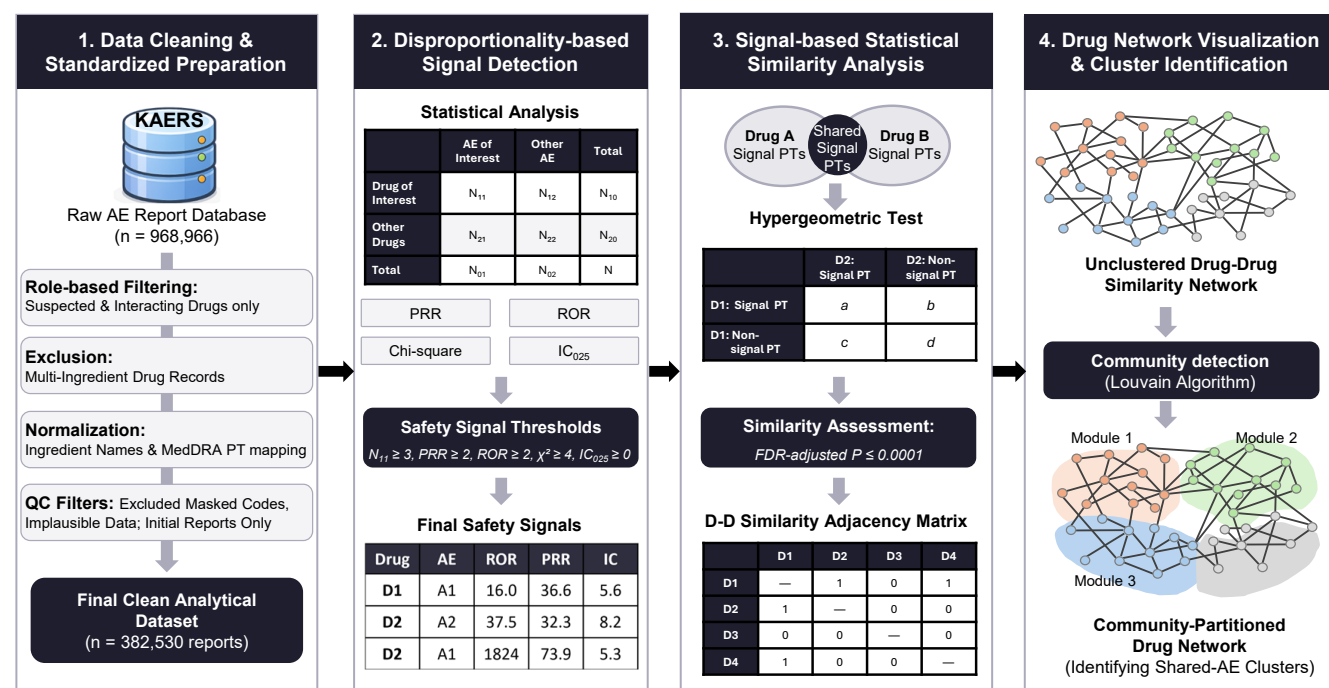


Fig. 1. Workflow for constructing a signal-based drug–drug similarity network using KAERS data. The pipeline involves (i) preprocessing of spontaneous reports, (ii) signal detection via disproportionality analysis, (iii) similarity quantification between ingredients based on overlapping safety signals using a hypergeometric test with FDR correction (BH-adjusted $p < 0.0001$), and (iv) network construction and community detection using the Louvain algorithm. KAERS, Korea Adverse Event Reporting System; FDR, false discovery rate; BH, Benjamini–Hochberg; AE, adverse event; PT, preferred term; PRR, proportional reporting ratio; ROR, reporting odds ratio; IC, information component; MedDRA, Medical Dictionary for Regulatory Activities.

generated using fully reproducible scripts.

RESULTS

Overview of the analytical workflow

Starting from 968,966 KAERS reports, data were systematically preprocessed to generate a curated analytical dataset comprising 382,530 reports, 1,058 unique ingredients, and 3,749 AEs (Fig. 1). Disproportionality analysis was then performed to identify statistically significant ingredient–AE associations using predefined multi-metric criteria (report count ≥ 3 , PRR ≥ 2 , ROR ≥ 2 , $\chi^2 \geq 4$, and $IC_{025} \geq 0$). This step yielded 10,711 significant ingredient–AE pairs involving 707 ingredients and 1,111 AEs. The optimal threshold for network construction was determined by evaluating network density across varying numbers of shared safety signals. A distinct inflection point was observed at 11 signals. While density reached a stable plateau at 12, this more stringent cutoff would have curtailed connectivity by 24% (to 959 edges), undermining the statistical power required for robust modular analysis. Consequently, the threshold of 11 was identified as the optimal balance, yielding a final sparse network of 150 ingredients and 1,267 edges. Finally, the community structure within the network was identified using the Louvain algorithm, which enabled the delineation of groups of drugs with convergent AE profiles.

Data preprocessing and analytic dataset construction

After preprocessing, the analytical dataset comprised 382,530 reports with standardized ingredients and AE annotations (Fig. 2). Preprocessing involved the exclusion of reports with implausible or incomplete information, multi-ingredient drug records, duplicate case entries, and non-systemic drug exposures. Preprocessing substantially altered the dataset composition (Table 1). Reports from consumers and non-medical reporters decreased markedly from 39.4% to 3.2%, whereas those from regional pharmacovigilance centers increased from 43.3% to 83.9%. Reports submitted by other medical professionals were the dominant category (61.5%), indicating the enrichment of clinically curated data sources.

The structural complexity of the reports was also reduced. The proportion of single-ingredient single-drug reports increased from 75.8% to 87.6%, whereas all multi-ingredient drug records were removed. Similarly, reports containing only a single AE increased from 56.7% to 70.6%, whereas multi-event reports decreased, reflecting reduced reporting complexity and the exclusion of duplicate or ambiguous entries. When individual drug records within reports were used as the unit of analysis (*i.e.*, report–drug pairs), the distribution of drug roles shifted markedly (Table 2). The proportion of report–drug pairs classified as suspected increased to 99.8%, whereas concomitant drug pairs were completely removed and interacting drug pairs were retained only minimally (0.21%). In contrast, demographic distributions at the report level remained largely stable, with a slight increase in older patients (≥ 61 years: 38.4% to 44.0%) and a consistent predomi-

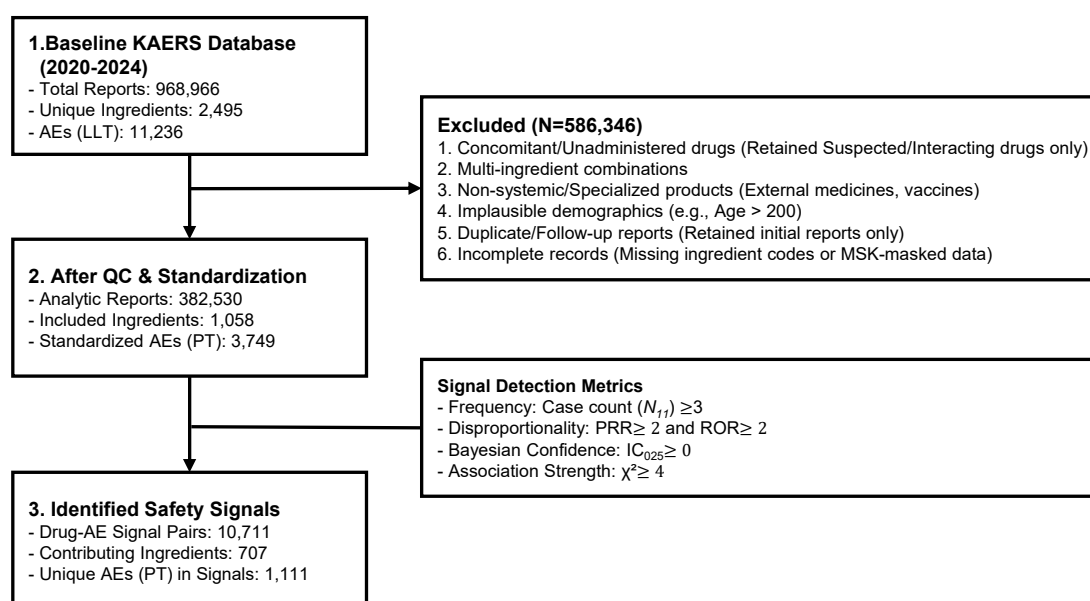


Fig. 2. Preprocessing and signal detection overview. Of the 968,966 raw reports, 382,530 reports were retained after exclusion of implausible, incomplete, or irrelevant entries. Safety signal detection was performed across all eligible ingredient–adverse events (AE) pairs using PRR, ROR, χ^2 , and IC_{025} criteria, yielding 10,711 safety signals across 707 ingredients and 1,111 AEs. KAERS, Korea Adverse Event Reporting System; LLT, lowest level term; PT, preferred term; PRR, proportional reporting ratio; ROR, reporting odds ratio; IC, information component.

Table 1. Report-level characteristics before and after preprocessing

Information of reports	Initial data (N = 968,966)	After preprocessing (N = 382,530)
Reporter qualification		
Consumers, non-medical professionals	382,010 (39.42)	12,316 (3.22)
Other medical professionals	309,421 (31.93)	235,257 (61.50)
Doctors, dentists, Korean medicine doctors	171,562 (17.71)	59,104 (15.45)
Pharmacists, Korean medicine pharmacists	90,424 (9.33)	67,512 (17.65)
Unknown	15,545 (1.60)	8,337 (2.18)
Lawyer	5 (0.0)	4 (0.0)
Reporting source		
Regional pharmacovigilance center	419,894 (43.33)	321,112 (83.94)
Regulatory authority	329,012 (33.95)	0 (0.0)
Pharmaceutical company	169,148 (17.46)	38,984 (10.19)
Healthcare professional	31,233 (3.22)	22,277 (5.82)
Other	19,539 (2.02)	80 (0.02)
Patient/consumer	141 (0.01)	77 (0.02)
Age groups (yr)		
0–7	11,381 (1.17)	4,308 (1.13)
8–20	44,343 (4.58)	13,851 (3.62)
21–40	211,437 (21.82)	64,123 (16.76)
41–60	330,114 (34.07)	131,828 (34.46)
61+	371,691 (38.36)	168,420 (44.03)
Sex		
Male	367,371 (37.91)	146,751 (38.36)
Female	588,009 (60.68)	230,273 (60.20)
Unknown	13,586 (1.40)	5,506 (1.44)
Role of drug per report ^a		
Suspected	968,386 (99.94)	382,098 (99.89)
Interacting	122,481 (12.64)	604 (0.16)
Concomitant	809 (0.08)	0 (0.0)
Other reports ^b	57 (0.01)	0 (0.0)
Drug and ingredient composition per report ^a		
Single-ingredient-single drug	732,489 (75.59)	335,177 (87.62)
Multiple-ingredient-single drug	48,989 (5.06)	0 (0.0)
Single-ingredient-multiple drug	98,428 (10.16)	47,353 (12.38)
Multiple-ingredient-multiple drug	2,390 (0.25)	0 (0.0)
Mixed (single- and multiple-ingredient drugs)	82,175 (8.48)	0 (0.0)
Other reports ^b	2,179 (0.22)	0 (0.0)
Number of lowest level terms per report		
1	549,878 (56.75)	270,127 (70.62)
2	208,360 (21.50)	82,972 (21.69)
3	111,372 (11.49)	20,410 (5.34)
4	41,452 (4.28)	5,501 (1.44)
≥ 5	57,814 (5.97)	3,520 (0.92)
Reports with fatal adverse events		
Serious ^c	6,727 (0.69)	1,273 (0.33)
Non-serious	962,239 (99.31)	381,257 (99.67)

Distribution of Korea Adverse Event Reporting System adverse event reports across characteristics before and after preprocessing, with reports as the unit of analysis. Each value represents the number of reports per category (n, %). Values are expressed as n (%) unless otherwise specified. ^aMultiple drugs can be reported per single adverse event report; ^bReports with missing drug information; ^cSerious adverse events include reports of death, life-threatening conditions, hospitalization, disability, or other clinically significant outcomes.

nance of female reports (~60%).

Collectively, these results reveal that preprocessing not only reduced the dataset size, but also reshaped the data toward a more interpretable structure by removing ambiguous drug exposures,

reducing reporting redundancy, and enriching clinically meaningful records (Tables 1 and 2).

Table 2. Drug-record characteristics before and after preprocessing

Information of drugs	Initial data (N = 1,989,253)	After preprocessing (N = 467,557)
Number of ingredient(s) per drug		
Single	1,783,823 (89.67)	467,557 (100.0)
Multiple	205,430 (10.33)	0 (0.0)
Role of drug		
Suspected	1,157,118 (58.17)	466,559 (99.79)
Interacting	1,700 (0.09)	998 (0.21)
Concomitant	830,378 (41.74)	0 (0.0)
Drugs without role information	57 (0.00)	0 (0.0)

Distribution of drug records across characteristics before and after preprocessing, with report–drug pairs as the unit of analysis. Each value represents the number of individual report–drug pairs (n, %) in each category, rather than the number of unique ingredients or drugs. The variables include the number of ingredients per drug record and the drug role.

Distribution of safety signals

Disproportionality analysis identified 10,711 ingredient–safety signals, involving 707 ingredients and 1,111 unique PTs (Fig. 2 and Supplementary Table 1). The safety signal burden varied markedly across ingredients. A small subset of ingredients accounted for a high frequency of distinct safety signals, including edoxaban (179 PTs), tacrolimus (170 PTs), aspirin (123 PTs), denosumab (105 PTs), and pitavastatin (93 PTs) (Fig. 3A). At the individual AE level, heartburn was the most prevalent PT, identified across 221 ingredients, followed by constipation (n = 175), sleeplessness (n = 163), and sleepiness (n = 161). These findings indicate that gastrointestinal and neurological symptoms were the most frequent safety signals among the 707 ingredients analyzed. These events largely correspond to common symptomatic complaints and are, therefore, likely to represent non-specific

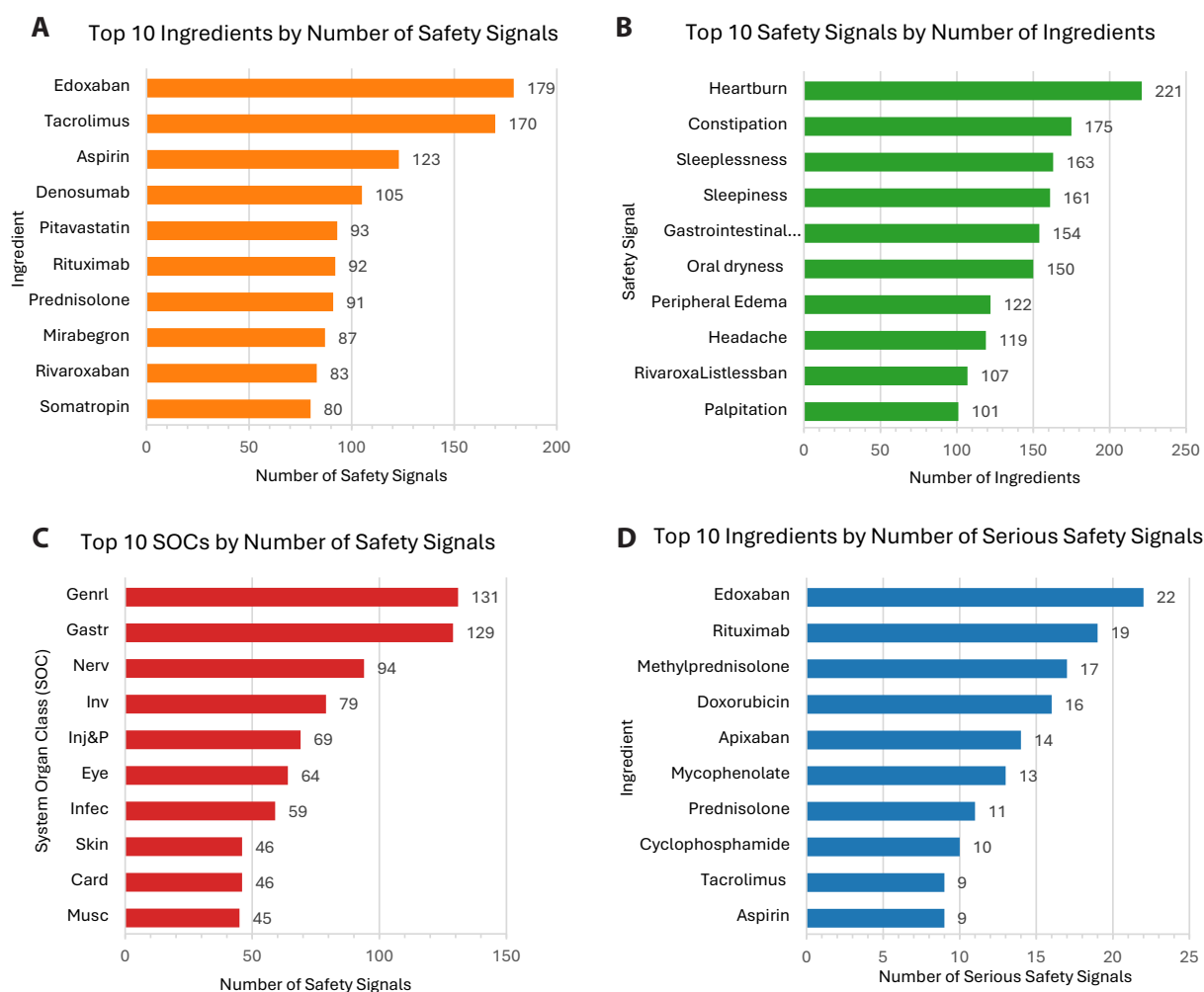


Fig. 3. Landscape of safety signal distribution. (A) Top 10 ingredients ranked by the number of safety signals. (B) Top 10 AEs ranked by the number of ingredients associated with them. (C) Distribution of AEs across MedDRA system organ classes (SOCs). (D) Top 10 ingredients with safety signals of serious AEs. AE, adverse event; MedDRA, Medical Dictionary for Regulatory Activities; Genrl, general disorders and administration site conditions; Gastr, gastrointestinal disorders; Nerv, nervous system disorders; Inv, investigations; Inj&P, injury, poisoning and procedural complications; Eye, eye disorders; Infec, infections and infestations; Card, cardiac disorders; Skin, skin and subcutaneous tissue disorders; Musc, musculoskeletal and connective tissue disorders.

reporting patterns rather than drug-specific toxicological effects, which was further corroborated by the MedDRA system organ class level analysis (Fig. 3C). Safety signals resulting in SAEs were rare, but were enriched in specific ingredients such as edoxaban and rituximab (Fig. 3D).

Signal-based drug–drug network construction

To compare the AE profiles across ingredients, we constructed a signal-based similarity network using shared safety signals. By applying the predefined criteria – specifically hypergeometric test ($FDR \leq 1 \times 10^{-4}$) and a minimum threshold of 11 shared safety signals – a robust network of associations was identified. The resulting network comprised 150 ingredients interconnected by 1,267 edges (Fig. 4A and Supplementary Table 2). The resulting

network showed a heterogeneous topology, with a small number of highly connected nodes and links observed both within and across ATC classes. Of these 1,267 edges, 1,114 (87.9%) linked ingredient pairs were co-reported in at least one KAERS report, whereas 153 pairs (12.1%) had no direct co-reporting history in the underlying dataset. The top-ranked ingredients by degree included rebamipide, mosapride, methylprednisolone, artemisia leaf extract, almagate, loxoprofen, sulpiride, gabapentin, pelubipirofen, and limaprost (Fig. 4B).

Community structure of the drug network

Community detection using the Louvain algorithm partitioned the network into 10 modules, with sizes ranging from 2 to 60 ingredients. Some modules were enriched in ingredients from the

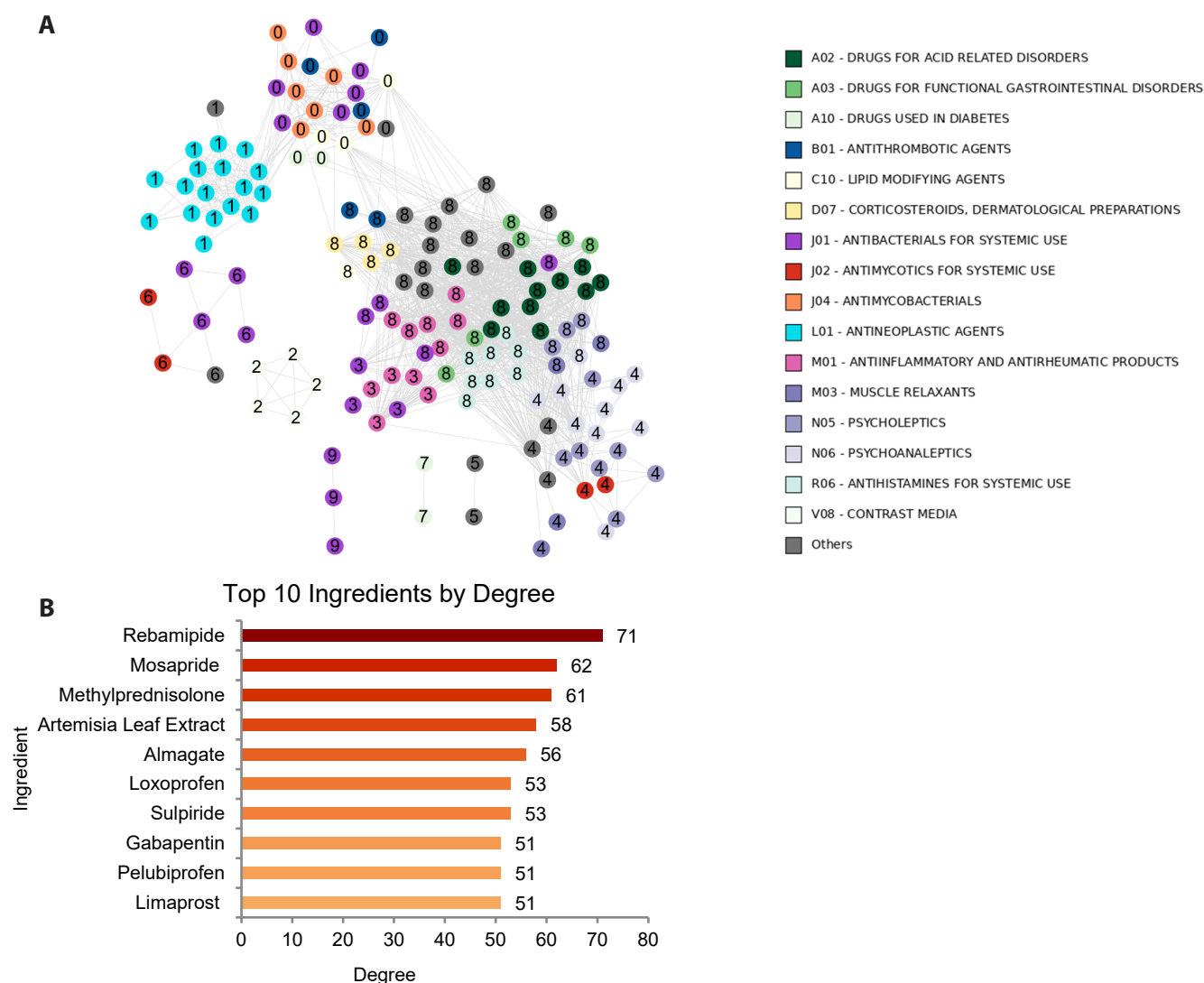


Fig. 4. Drug network based on shared safety signals. (A) Nodes represent ingredients; edges represent ingredient pairs sharing ≥ 11 safety signals ($FDR < 0.0001$). Node color indicates the most frequent second-level ATC classification. Module numbers correspond to Louvain-inferred communities. (B) Top 10 network hubs ranked by degree. Degree reflects the number of connected ingredients that share ≥ 11 safety signals with each node. High-degree drugs show broad signal overlap with diverse pharmacologic classes. FDR, false discovery rate; ATC, anatomical therapeutic chemical.

same ATC class, whereas others comprised ingredients from multiple ATC classes (Fig. 4A), indicating a variable correspondence between the network modules and therapeutic classification.

Among these, Module 1 showed the most homogeneous composition (Fig. 5A). This module comprised 18 ingredients, of which 17 were assigned to ATC L01 (antineoplastic agents), and were associated with hematological AEs, including leukopenia, neutropenia, and decreased neutrophil count (Supplementary Table 2). Within this module, cyclophosphamide, doxorubicin, and rituximab shared more than 30 safety signals and were also frequently co-reported within the same KAERS reports (shared KAERS reports > 600), indicating that their network proximity was supported by both signal overlap and recurrent co-reporting. In contrast, paclitaxel and docetaxel, both taxane antineoplastic agents with a shared microtubule-stabilizing mechanism, shared 25 significant signals despite no co-reported KAERS reports. Their network proximity was therefore not attributable to report-level co-occurrence.

Module 2 formed another relatively homogeneous cluster, consisting of contrast media and was associated primarily with

hypersensitivity-related events such as itching and urticaria (Fig. 4A and Supplementary Table 2). The other modules showed more specific event profiles, including hypersensitivity and edema (Module 3), rash and hepatic abnormalities (Module 6), nervous system-related symptoms (Module 4), injection-site reactions (Module 5), and diabetes-related profiles (Module 7).

In contrast, Module 0 comprised 22 ingredients and showed a densely connected structure spanning multiple ATC classes (Fig. 5B). Within this module, several pairs with shared pharmacological mechanisms showed strong connectivity, despite the absence of report-level co-occurrence. For example, rivaroxaban and edoxaban, both direct oral anticoagulants, shared 64 significant signals despite only two co-reported KAERS reports. Similarly, statins (atorvastatin, rosuvastatin, and pitavastatin) shared more than 30 significant signals with one another, while being co-reported in fewer than five reports.

Notably, multiple statin-anti-tuberculosis drug pairs, including one statin (atorvastatin, rosuvastatin, or pitavastatin) and one anti-tuberculosis agent (pyrazinamide, cycloserine, kanamycin, linezolid, or prothionamide), shared at least 15 safety signals de-

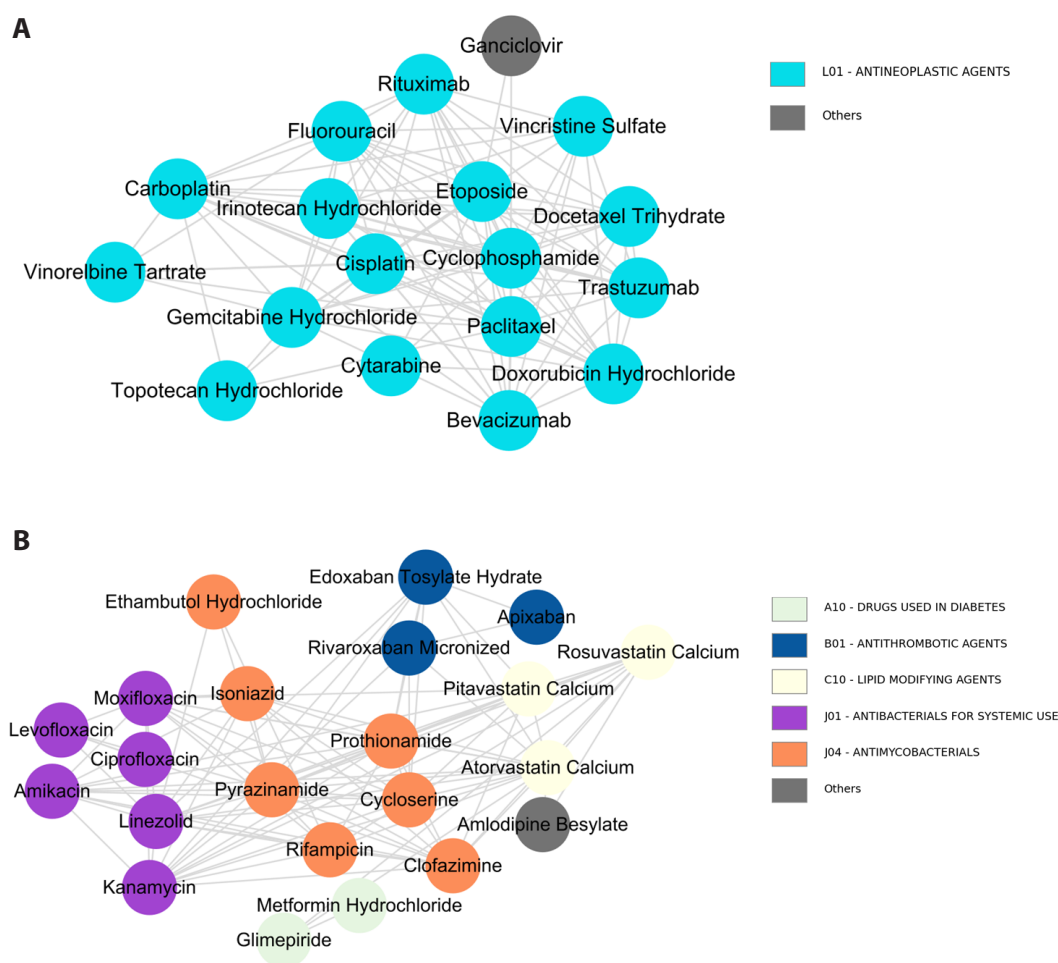


Fig. 5. Subnetworks of Modules 1 and 0 in the drug network. (A) Subnetwork of Module 1. (B) Subnetwork of Module 0. Nodes represent drug ingredients, and edges represent ingredient pairs sharing ≥ 11 safety signals with $\text{FDR} \leq 1 \times 10^{-4}$. Node colors indicate the predominant ATC second-level class assigned to each ingredient. FDR, false discovery rate; ATC, anatomical therapeutic chemical.

spite no co-reported KAERS reports (Fig. 5B and Supplementary Table 2). These cross-ATC links were primarily characterized primarily by signals related to hepatic dysfunction (e.g., aspartate aminotransferase/alanine aminotransferase increased and abnormal liver function test results), musculoskeletal symptoms (e.g., arthralgia, myalgia, and back pain), gastrointestinal complaints (e.g., gastrointestinal distress, heartburn, and gastritis), and neuropsychiatric symptoms (e.g., memory loss, depression, sleeplessness, and paresthesia). This pattern may be relevant for targeted safety monitoring when the two drug classes are used sequentially or concomitantly.

DISCUSSION

This study represents a methodological contribution in pharmacovigilance by transitioning from atomistic drug-event pairs to a holistic, ingredient-level drug-drug network utilizing the KAERS database. By converting thousands of ingredient-event pairs into a sparse similarity graph, our framework provides a view complementary view to conventional safety-signal tables. The primary strength of this approach lies in its ability to contextualize ingredients within their neighborhood of drugs with overlapping, statistically supported events, thereby facilitating the interpretation of clinically coherent summary of predominant phenotypes. The methodological robustness of our network is supported by the high degree of intra-class cohesion observed across drug classes. Without any prior pharmacological labeling, our data-driven algorithm identified clusters that align with established therapeutic classes—most notably contrast media, non-steroidal anti-inflammatory drugs (NSAIDs), antineoplastics, psychotropics, and antiepileptics—based solely on shared AE signatures (Fig. 4 and Supplementary Table 2). This recovery of well-recognized class-specific liabilities suggests that the network effectively captures underlying pharmacological patterns and clinical associations inherent in the KAERS database. For instance, the antineoplastic drugs were characterized by a clinically consistent cluster of hematological toxicities—including neutropenia, thrombocytopenia, febrile neutropenia, and anemia—alongside systemic manifestations such as stomatitis and alopecia. Similarly, the psychotropic drugs were associated with a signature of extrapyramidal symptoms, including tardive dyskinesia, akathisia, and Parkinsonism, as well as metabolic and endocrine markers such as weight increase and blood prolactin elevation. These findings provide face validity for our network construction strategy, suggesting that the framework effectively captures established pharmacological profiles from spontaneously reported data.

The analysis also identified the emergence of the antiepileptic drug (AED) class that exhibited a dual-mode safety signature. Unlike the antineoplastic module's focus on hematological toxicity, the AED cluster captured two distinct clinical phenotypes:

(1) Central nervous system effects characterized by dizziness, somnolence, ataxia, and blurred vision; and (2) idiosyncratic immunological liabilities, represented by severe cutaneous adverse reactions such as Stevens-Johnson syndrome, toxic epidermal necrolysis, and drug reaction with eosinophilia and systemic symptoms syndrome. Despite the disparate molecular mechanisms of action (MOAs) among individual agents, the high connectivity within these drug classes indicates that the network is highly sensitive to the clinical phenotype of the treated population. These findings indicate that the network approach is capable of capturing the clinical patterns and the AE burden observed in real-world practice.

Against this backdrop of high in-class reliability, the most notable observation of our analysis was the convergence of statins, specifically pitavastatin, atorvastatin, and rosuvastatin, with a cluster of ingredients traditionally characterized by their anti-inflammatory and anti-mycobacterial properties (e.g., linezolid, clofazimine, and pyrazinamide). Although statins are primarily HMG-CoA reductase inhibitors, their pleiotropic effects, including the modulation of inflammatory pathways and C-reactive protein reduction, are well documented in the literature [19–23]. While the majority of the network reinforces known therapeutic silos, this particular cross-class convergence suggests that our framework may capture latent signals of systemic pharmacological modulation. This clustering indicates that beyond their primary lipid-lowering indication, statins may produce a safety signature that overlaps with that of systemic anti-inflammatory agents, reflecting common downstream biological pathways or shared metabolic vulnerabilities. This finding suggests that the developed KAERS-derived network can be useful to transcend primary therapeutic indications to elucidate shared toxicological pathways across disparate therapeutic classes.

The high connectivity observed among gastrointestinal agents and NSAIDs is consistent with established co-prescription patterns in clinical practice. The frequent concurrent reporting of these drug classes as suspected agents likely contributes to the statistical overlap of their AE signatures, resulting in a high density of network edges (Fig. 4A). Conversely, among ingredients outside of the high-volume therapeutic classes, methylprednisolone emerged as a notable hub. This positioning potentially reflects its systemic safety profile, which encompasses effects across multiple organ systems and facilitates linkage with diverse modules. Additionally, the extensive signature similarities demonstrated by sulpiride and gabapentin across disparate therapeutic areas are attributable to their pleiotropic MOAs and usage across multiple clinical indications.

Despite the clinical coherence of our network, it is subject to the intrinsic limitations of spontaneous reporting systems. KAERS lacks explicit exposure denominators and complete treatment histories, precluding incidence estimation and limiting our interpretation to reporting-based associations rather than causal effects [24]. Furthermore, reporting behavior is highly suscep-

tible to external factors—such as media attention or regulatory actions—which can induce stimulated reporting and affect both signal strength and network connectivity. Clinical confounding by indication, disease severity, and polypharmacy also remains difficult to fully address using report-level summaries alone, as co-reported symptoms may reflect the underlying illness rather than the true drug effects.

The potential influence of prescription volume warrants consideration, as a greater number of reports typically enhances statistical power for signal detection. However, our framework identifies clusters based on the specific composition of overlapping safety signals rather than absolute counts, with hypergeometric testing and FDR correction ensuring that edges represent significant pattern-based concordance. While this approach is less susceptible to volume-related bias, it remains subject to certain methodological constraints. For instance, the Louvain algorithm introduces a resolution limit, a known phenomenon where smaller, clinically distinct sub-modules are forced into larger, more heterogeneous communities to optimize global modularity [25]. This likely explains the emergence of the symptom-heavy Module 8, which may act as a sink for non-specific reporting noise. Additionally, our choice of binary encoding and stringent edge thresholds (≥ 11 shared safety signals), while reinforcing face validity, may have excluded drugs with narrow but clinically profound safety profiles.

Another limitation of this study is the reliance on the ATC 2nd-level taxonomy for module annotation, which may not consistently reflect actual clinical utility. For example, within the ATC hierarchy, Ginkgo biloba is categorized as a psychostimulant (N06B), and certain agents primarily used for neuropathic pain are classified as analgesics (N02B) rather than antiepileptics. However, this discrepancy underscores a key strength: because the network construction was blinded to administrative labels, the algorithm grouped ingredients solely based on shared AE signatures, identifying functional phenotypes that exist independently of traditional taxonomic systems. Finally, excluding vaccines (primarily reported by the KDCA) and non-systemic/multi-ingredient drugs resulted in the complete exclusion of regulatory authority reports (from 33.95% to 0%) and a proportional increase in regional pharmacovigilance center reports (from 43.33% to 83.94%). While this refined, professional-centric subset maximizes signal clarity, it may limit the generalizability of our findings compared to the broader source database.

Building upon these findings, future investigations could specifically focus on constructing drug similarity networks based on SAEs. While the current study provides a comprehensive overview of the broader safety signal landscape, a targeted analysis of SAEs—which carry greater regulatory weight and clinical urgency—may reveal high-risk drug clusters that differ from a generalized network. Such a focused approach would provide more critical insights for regulatory surveillance and the refinement of risk-management strategies.

In conclusion, this study demonstrates that an ingredient-level integration of KAERS signals into a drug–drug network can help characterize clinically coherent phenotypes and serve as an exploratory hypothesis-generation scaffold. By contextualizing drugs through their shared AE signatures, our framework offers a potential approach for signal prioritization and aids in the identification of cross-therapeutic similarities that might be overlooked by conventional class-based summaries. In the meanwhile, these findings should be interpreted as preliminary associations, given the inherent limitations of spontaneous reporting data. Future research should leverage this network to prioritize these identified clusters for evaluation in longitudinal healthcare data, such as electronic health records or insurance claims, to further validate their clinical relevance.

FUNDING

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (Nos. RS-2024-00440285, RS-2024-00432867, and RS-2025-02283193); the Korea Institute of Oriental Medicine (Grant number KSN2235120); and the Dongnam Institute of Radiological & Medical Sciences (DIRAMS) grant funded by the Korea government (MSIT) (No.50703–2026).

ACKNOWLEDGEMENTS

None.

CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

SUPPLEMENTARY MATERIALS

Supplementary data including two tables can be found with this article online at <https://doi.org/10.4196/kjpp.26.080>

REFERENCES

1. Dal Pan GJ, Lindquist M, Gelperin K. Postmarketing spontaneous pharmacovigilance reporting systems. In: Strom BL, Kimmel SE, Hennessy S, editors. *Pharmacoepidemiology*. Wiley; 2019.
2. Fang H, Su Z, Wang Y, Miller A, Liu Z, Howard PC, Tong W, Lin SM. Exploring the FDA adverse event reporting system to generate hypotheses for monitoring of disease characteristics. *Clin Pharmacol Ther*. 2014;95:496–498.
3. Bate A, Evans SJ. Quantitative signal detection using spontaneous

- ADR reporting. *Pharmacoepidemiol Drug Saf.* 2009;18:427-436.
4. Soukavong M, Kim J, Park K, Yang BR, Lee J, Jin XM, Park BJ. Signal detection of adverse drug reaction of amoxicillin using the Korea adverse event reporting system database. *J Korean Med Sci.* 2016;31:1355-1361.
 5. Chai S, Zhan JL, Zhao LM, Liu XD. Safety of triazole antifungals: a pharmacovigilance study from 2004 to 2021 based on FAERS. *Ther Adv Drug Saf.* 2022;13:20420986221143266.
 6. Huang J, Zhang M, Lin J, Yang X, Huang P, Zheng X. Safety assessment of proteasome inhibitors real world adverse event analysis from the FAERS database. *Sci Rep.* 2025;15:11628.
 7. Liu H, Yang Q, Li Z, Yan S, Ming S. Systematic analysis of sugammadex-related adverse drug reaction signals using FAERS database. *Int J Surg.* 2025;111:1988-1994.
 8. Liu D, Li N, Liu L, Qiu L, Zhou Z, Wu J. Uncovering safety signals of duloxetine: real-world evidence from the FAERS database. *J Affect Disord.* 2025;388:119517.
 9. Lv L, Wu X, Ren Y, Guo Y, Wang H, Li X. Postmarketing safety evaluation of pemetrexed using FAERS and JADER databases. *Sci Rep.* 2025;15:18738.
 10. Milašinović B, Vezmar Kovačević S, Marković S, Jovanović M, Knežević Ivanovski T, Kralj Đ, Svorcan P, Miljković B, Vučićević K. Real-world safety of vedolizumab in inflammatory bowel disease: a retrospective cohort study supported by FAERS signal analysis. *Pharmaceuticals (Basel).* 2025;18:1127.
 11. Tang X, Wang C, Li Y, Tang J, Zhang G, Chen L. Signal detection and safety analysis of three tyrosine kinase inhibitors for HER-2 positive breast cancer: a retrospective study based on the FAERS database. *Front Pharmacol.* 2025;16:1538881.
 12. Yang F, Xu L, Lang X, Feng X. A real-world safety signal detection study of ondansetron based on FAERS reports from 2014 to 2024. *Sci Rep.* 2025;15:31339.
 13. Ye G, Ding L, Zhou Q. Remimazolam's clinical application and safety: a signal detection analysis based on FAERS data and literature support. *PLoS One.* 2025;20:e0330769.
 14. Ye W, Ding Y, Li M, Tian Z, Wang S, Liu Z. Safety assessment of sulfasalazine: a pharmacovigilance study based on FAERS database. *Front Pharmacol.* 2024;15:1452300.
 15. Ye H, Liu Q, Wei J. Construction of drug network based on side effects and its application for drug repositioning. *PLoS One.* 2014;9:e87864.
 16. Korea Institute of Drug Safety & Risk Management. Introduction of KAERS [Internet]. [cited 2026 Feb 1]. Available from: <https://www.drugsafe.or.kr/iwt/ds/en/report/WhatIsKAERS.do>
 17. Newman M. Mathematics of networks. In: Newman M, editor. Networks. Oxford University Press; 2018. p.128-130.
 18. Blondel VD, Guillaume J, Lambiotte R, Lefebvre E. Fast unfolding of communities in large networks. *J Stat Mech.* 2008;2008:P10008.
 19. Blake GJ, Ridker PM. Are statins anti-inflammatory? *Curr Control Trials Cardiovasc Med.* 2000;1:161-165.
 20. Ridker PM, Cannon CP, Morrow D, Rifai N, Rose LM, McCabe CH, Pfeffer MA, Braunwald E. C-reactive protein levels and outcomes after statin therapy. *N Engl J Med.* 2005;352:20-28.
 21. Blanco-Colio LM, Tuñón J, Martín-Ventura JL, Egido J. Anti-inflammatory and immunomodulatory effects of statins. *Kidney Int.* 2003;63:12-23.
 22. Taqueti VR, Ridker PM. Lipid-lowering and anti-inflammatory benefits of statin therapy: more than meets the plaque. *Circ Cardiovasc Imaging.* 2017;10:e006676.
 23. He WB, Ko HTK, Curtis AJ, Zoungas S, Woods RL, Tonkin A, Neumann JT, Turner SL, Hopper I. The effects of statins on cardiovascular and inflammatory biomarkers in primary prevention: a systematic review and meta-analysis. *Heart Lung Circ.* 2023;32:938-948.
 24. Dhodapkar MM, Shi X, Ramachandran R, Chen EM, Wallach JD, Ross JS. Characterization and corroboration of safety signals identified from the US Food and Drug Administration Adverse Event Reporting System, 2008-19: cross sectional study. *BMJ.* 2022;379:e071752.
 25. Fortunato S, Barthélemy M. Resolution limit in community detection. *Proc Natl Acad Sci U S A.* 2007;104:36-41.